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22. Assertion : Critical temperature of CO_2 is 304K, it can be liquefied above 304K.

Reason : For a given mass of gas, volume is to directly proportional to pressure at constant temperature

a) both assertion and reason are true and reason is the correct explanation of assertion

b) both assertion and reason are true but reason is not the correct explanation of assertion

c) assertion is true but reason is false

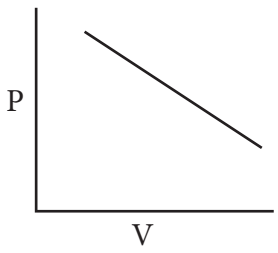
d) both assertion and reason are false

23. What is the density of N_2 gas at 227°C and 5.00 atm pressure? (R = 0.082 L atm K–1 mol^{-1})

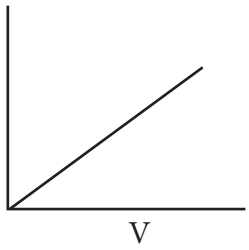
a) 1.40 g/L b) 2.81 g/L c) 3.41 g/L d) 0.29 g/L

24. Which of the following diagrams correctly describes the behaviour of a fixed mass of an ideal gas ? (T is measured in K)

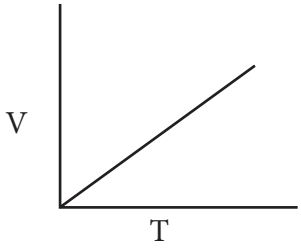
a)



b) PV



c)



d) All of these

25. 25g of each of the following gases are taken at 27°C and 600 mm Hg pressure. Which of these will have the least volume ?

a) HBr b) HCl c) HF d) HI